

KrishokTech: Deterministic-First, Evidence-Bounded Bengali Agricultural Advisory

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Abstract

We present KRISHOKTECH, a Bengali agricultural advisory system designed for practical use in Bangladesh under limited connectivity, device, and knowledge-resource conditions. Rather than treating a language model as the decision maker, KRISHOKTECH uses a deterministic-first pipeline that can halt underspecified treatment queries before retrieval, constrain retrieval to the crop, expose image-text contradictions to the user, and verify safety-critical dosage claims before they are rendered or delivered. The system also supports constrained offline and SMS delivery and provides a terminal path to human agricultural assistance for cases that should not be answered automatically. We demonstrate the system through realistic Bengali farmer interactions covering missing crop information, image-based crop routing, contradictory evidence, grounded responses, unsafe requests, and constrained delivery. Evaluation shows that the pre-retrieval gate halted 30/200 treatment queries, while crop fencing reduced wrong-crop answers from 36.25% to 30.0% in a controlled evaluation, with 25 discordant cases favoring the fenced setting. Across 420 live safety calls in the cloud-model benchmark, guarded responses reduced attack success rate from 36.19% to 0.95%. On the deployed local model the same guard cuts attack success from 52.5% to 10.71% across 560 calls (Bangla-100: 77.0% to 19.0%). These results illustrate how explicit constraints can make agricultural conversational systems more controllable without relying solely on a larger language model.

1 Introduction

For a farmer in Bangladesh, agricultural advice is urgently needed when symptoms appear, pests spread, or treatment cannot wait for an extension visit. Agriculture accounted for 11.7% of GDP and 44.3% of employment in 2025 (World Bank, 2026), yet smallholders face a severe extension

workforce deficit (Department of Agricultural Extension, 2022; Developing Local Extension Capacity (DLEC) Project, 2021). A digital assistant can extend access, but only if it accurately recognizes when requests are underspecified or unsafe.

The challenge goes beyond providing a Bengali chat interface. Farmers describe symptoms in colloquial or regional Bangla, whereas agricultural resources use standardized terms; our prior work found a substantial formal-to-colloquial retrieval gap (Reza et al., 2026a). Retrieval can also fail before generation: treatment requests often omit the crop, photographs may contradict text, and generated dosages may lack evidence grounding. Fluent models turn unresolved uncertainties into confident, dangerous recommendations. Prior work revealed residual unsupported chemical advice even under oracle evidence (Reza et al., 2026b), and vision-language models often exhibit blind faith in contradictory text (Deng et al., 2025).

We present **KrishokTech**, a Bengali agricultural advisory system structured around explicit decision boundaries rather than ungrounded generation: First, *the cheapest and safest retrieval is the one the system refuses to run (C1)*: when a treatment query lacks a crop, a deterministic gate halts before search and serves clarification chips or helpline referral (halting 30/200 farmer queries with zero retrieval). Second, *the photo is a fence around the evidence (C2)*: on-device vision bounds retrieval to the identified crop; text-image conflicts raise an explicit confirmation badge (reducing wrong-crop advice from 36.25% to 30.0% in gold-routing simulation; catching 453/454 explicit contradictions). Third, *the LLM is the mouthpiece; the evidence is the witness; the verifier is the judge (C3)*: generated dosage claims remain untrusted until verified against supporting passages and permitted dose bands. Unsupported claims drop at render time across web, SMS, offline cards, or 16123 referral (cloud-model benchmark guarded

ASR 0.95% vs 36.19%; verifier 114/118; deployed local model 52.5% to 10.71%). Failure handling is thus a visible interaction state. Interactive demo: <https://krishoktech.org> (screencast, <2.5 min: <https://krishoktech.org/screencast>); source code and installable package (Apache-2.0): <https://github.com/RaiyaanReza/KrishokTech-Agricultural-Advisory-System>.

2 Related Work and Positioning

Conversational agricultural advisory systems have demonstrated the value of combining localized agricultural knowledge with conversational interfaces. Farmer.Chat provides multilingual, multimodal assistance via curated RAG and structured crop onboarding (Singh et al., 2024). KrishokBondhu targets Bengali farmers through a voice-first call-centre interface combining speech recognition, retrieval, and generation (Ameen et al., 2026). Krishi Sathi uses intent-aware multi-turn slot collection before retrieval (Vijayvargia et al., 2025), while Bengali RAG uses translation and keyword injection to bridge colloquial and formal terms (Hossain et al., 2026). These systems establish conversational access, retrieval grounding, multimodal interaction, and clarification as foundational ingredients.

A second line of work focuses on evidence grounding and output safety. My Climate CoPilot exposes intermediate tool traces and uses expert post-hoc self-evaluation (Nguyen et al., 2025). AgroLLM incorporates structured agro-economic thresholds to validate outputs (Ravindran et al., 2026). In multimodal settings, vision-language models frequently favor contradictory text over visual evidence (Deng et al., 2025). KrishokTech addresses an operational challenge: composing these mechanisms into enforceable, visible control states within a unified Bengali advisory flow. Rather than treating components in isolation, the system operationalizes pre-answer stopping, crop-conditioned retrieval fencing, pre-render dosage verification, and terminal human referral in a farmer-facing workflow (see comparison below).

Mechanism / Decision	Ours	FC	KB	KS	MC
Pre-retrieval halt on missing crop	Yes	No	No	Part	No
Visual crop-fenced retrieval	Yes	No	No	No	No
Pre-render dosage-claim verifier	Yes	No	No	No	Part
Terminal crisis/poisoning referral	Yes	No	Part	No	No

System comparison across control mechanisms based on capabilities documented in cited system descriptions: Farmer.Chat (FC), KrishokBondhu (KB), Krishi Sathi (KS), and My Climate CoPilot (MC).

3 System Design

KrishokTech organizes the advisory process around four successive decision boundaries rather than a single retrieve-and-generate step (Figure 1). Each boundary narrows what the next stage may do: **T0** screens raw requests for unsafe content; **T1** decides whether the request carries enough information to search; **T2** constrains what evidence search may return; and **T3–T4** treat generator output as an unverified draft.

3.1 Halting Before Retrieval

At **T0**, a deterministic precheck matches incoming Bengali queries against banned chemicals (e.g., paraquat), poisoning, and crisis patterns in 0.32 ms, routing immediately to the national 16123 helpline with zero LLM invocation. Requests clearing **T0** reach **T1**, where a deterministic extractor pulls crop, symptom, and treatment intent. When a treatment query lacks a resolvable crop, KrishokTech treats this as a missing safety-critical slot. Instead of unconstrained search, the pipeline halts before BM25 runs (*sources_retrieved* = 0) and offers three bounded outcomes: a non-chemical guidance card, interactive clarification with Bengali quick-reply crop chips, or 16123 referral. A *Dialect-Selector* provides six regional presets to bridge the colloquial-to-formal retrieval gap (Reza et al., 2026a).

3.2 Crop-Fenced Retrieval and the Mismatch Badge

Requests clearing **T1** enter **T2**. KrishokTech retrieves via BM25 over 2,135 curated BARI, BRRI, and DAE guideline nodes (Bangladesh Agricultural Research Institute, 2023; Bangladesh Rice Research Institute, 2024; Department of Agricultural Extension, 2022), but only after crop identity is bound. The current deployment is optimized for six staple crops (rice, potato, brassica, chili, corn, wheat) noted unobtrusively in the interface. For photographs, an on-device ONNX vision model classifies the crop; a tri-state router directs confident predictions to the species manual (State A), ambiguous pairs to disambiguation chips (State B), and out-of-scope inputs to image re-capture (State C). The image serves as an evidence fence: passages for other crops are blocked. When text and image contradict each other, KrishokTech raises an active **Crop Mismatch** badge, holding chemical advice until the farmer confirms the crop. If re-

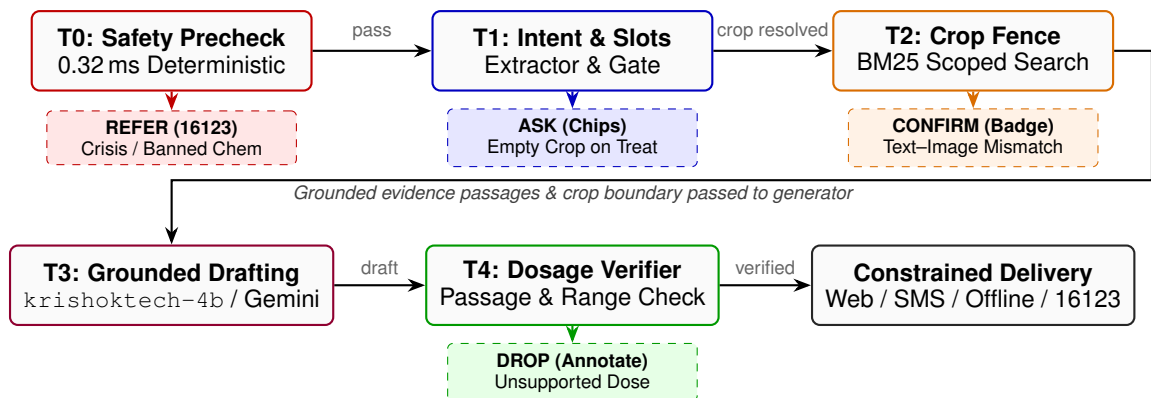


Figure 1: The complete KrishokTech five-stage decision flow in a two-row pipeline layout. Execution begins at T0 with deterministic safety screening and advances through intent/slot extraction (T1), crop-fenced BM25 retrieval (T2), untrusted language model drafting (T3), and pre-render dosage verification (T4). Four colored exit gates halt execution or alter delivery: REFER to 16123 helpline at T0, ASK clarification chips at T1, CONFIRM mismatch badge at T2, and DROP unsupported dosage claims at T4.

trieved evidence within a crop contains conflicting pathogen diagnoses, the system asks a discriminative clarification question rather than blending contradictory treatments.

3.3 Untrusted Generation, Verification, and Delivery

At **T3**, *krishoktech-4b* (fine-tuned 4-bit quantized Gemma base model with a LoRA adapter on *llama-server*) performs grounded drafting, with *google/gemini-2.5-flash-lite* as the cloud benchmark. The draft is treated as untrusted. At **T4**, the dosage verifier checks whether numerical claims are bound to a retrieved passage and fall within permitted BARI/BIRRI dose bands; failing claims are annotated and dropped. Surviving responses carry a *provenance badge* (fact lookup, grounded generation, guidance card, clarification, or referral). The system renders the verified decision across four channels: interactive web advisory, 160-character SMS, offline cached cards via local BM25, or sentence-synchronized Bengali neural read-aloud.

4 Demonstration

The demonstration targets NLP researchers and practitioners in multilingual RAG, multimodal conversational systems, and safety-critical dialogue. Attendees interact with the live system across its five operational states: submitting colloquial Bengali to observe deterministic pre-search halting, testing quick-reply disambiguation chips, uploading conflicting leaf images to trigger the contradiction badge, tracing verified dosage citations, and

observing fail-closed helpline referral. The session proceeds through one continuous interaction:

S1: Halt, clarify, resume. The farmer opens with a colloquial symptom query that names no crop (“*Patay holud dag hoyeche, ki bish dibo?*” / “*There are yellow spots on the leaves, which pesticide should I apply?*”). KrishokTech treats the missing crop as a safety-critical omission and halts before search (*sources_retrieved* = 0), presenting a short Bengali clarification with three tappable crop chips. Tapping *Alu* (Potato) binds the crop and resumes retrieval. The interface demonstrates that the system asks before searching, allowing user selection to scope admissible evidence.

S2: A photograph narrows the evidence. The farmer follows up with a leaf photograph. On-device image classification identifies it as potato, and this identification, not the text alone, sets the boundary of what the retrieval step may return: only potato guidance is reachable from this point on. The farmer sees the identified crop and evidence that is visibly scoped to it, not a caption passed silently to a language model.

S3: The photo and the words disagree. Later in the same exchange, the farmer types a question about rice blast while the potato photograph from S2 is still attached. KrishokTech does not pick a side. A **Crop Mismatch** badge appears and chemical guidance is held until the farmer confirms which crop is meant.

S4: A request the system will not answer. In the same session, the farmer submits a request naming a banned active ingredient. KrishokTech recognizes this before retrieval or generation and stops

the turn entirely. No advice is generated as a softer alternative; the interface shows a direct referral to the national Krishi Call Centre (16123).

S5: A grounded answer, checked before delivery. The farmer returns to a fully specified question. KrishokTech now proceeds to generation: it drafts a Bengali advisory and checks every dosage-bearing sentence against a retrieved passage and a permitted dose range before showing the answer. In the demonstrated case, one dosage sentence fails this check and is dropped; the farmer sees the surviving, verified answer with its source citations and a provenance badge, and can open the Why/trace panel to see exactly which sentence was removed and why (Figure 2). The same checked answer can then be delivered as a length-constrained SMS or an offline cached card, with the interface making clear that exact dosage figures must come from the 16123 helpline once truncation shortens the message.

Across S1–S5, the same principle recurs from a different angle each time: KrishokTech does not treat every farmer message as a generation prompt. It can stop before search, let a photograph bound what evidence is admissible, surface disagreement between modalities instead of resolving it silently, end an unsafe request in a human referral, and check a generated claim before the farmer ever reads it.

5 Evaluation

The preceding section demonstrated how KrishokTech handles underspecified, multimodal, unsafe, and constrained interactions. We now evaluate whether the corresponding decision boundaries behave as intended under the measured test conditions. The evaluation is organized around the three core contributions: whether the system halts underspecified treatment requests before retrieval (C1), whether crop information consistently constrains evidence and catches text–image contradictions (C2), and whether generated dosage claims remain verified and safe across constrained delivery channels (C3). Table 1 establishes the empirical status and validated scope of all headline claims (a comprehensive evidence breakdown across all pipeline layers is provided in Table 13, Appendix G).

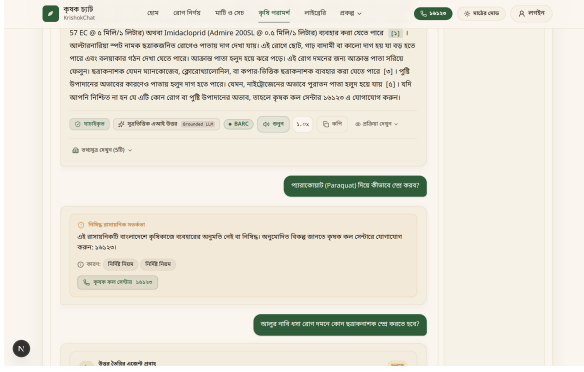
5.1 C1: Pre-Retrieval Gating

We evaluate the deterministic gate on 200 authentic Bengali farmer treatment queries (Table 1). The

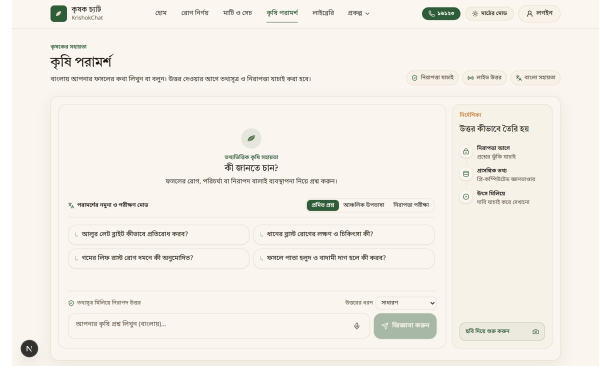
gate halted all 30 crop-less symptom requests before retrieval (*sources_retrieved* = 0), passing 170 crop-specified queries to fenced search (100% gate decision accuracy). Halted turns retrieved zero sources, preventing ungrounded generation. Ambiguity safely yields null extraction to trigger interactive quick-reply chips rather than ungrounded speculation; ground-truth slots were verified by independent dual annotation (100% binary agreement, $\kappa = 1.0000$; 99.5% taxonomy, $\kappa = 0.9945$). Across the 170 specified queries, slot extraction achieved an agreement of 95.5% (with genuine crop mentions recognized across multi-crop and sourcing intents). A pilot check confirmed the mechanism: 100/100 ambiguous requests halted; 114/140 specified passed directly, while 26/140 prompted bounded slot clarification. Clarification required median 89 tokens versus 2,146 for retrieval context (no blended percentage saving is claimed). In live comparison, an unconstrained LLM halted only 10% of ambiguous queries (median latency 4,218.6 ms; p95: 6,645.1 ms).

5.2 C2: Crop Constraint and Multimodal Consistency

Top-1 accuracy across all six crop-specific disease models was 0.9504 (potato, $n = 1,170$), 0.9625 (rice, $n = 80$), 0.9413 (wheat, $n = 800$), 0.9729 (brassica, $n = 443$), 0.9723 (corn, $n = 940$), and 0.9907 (chilli, $n = 861$), totaling 4,294 evaluated test images with 100% ONNX-to-PyTorch prediction agreement within 5×10^{-5} tolerance. Each model diagnoses diseases within its crop; species identity is established upstream by a frozen botanical family mapping (433/437 correct, 0.9908 [97.67, 99.64]; wheat-800 excluded). Static INT8 quantization reduces model weights by $3.68\times$ (from 5.92 MB to 1.61 MB for wheat, 0.00 pp drop; 0.09 pp drop on potato; 0.00 pp on brassica), though on non-VNNI x86 CPUs INT8 latency is $1.5\text{--}2.0\times$ higher than FP32 across 1–4 threads due to software dot-product emulation. On a 400-query stress benchmark of dialectal, ambiguous, and conflicting queries ($n = 400$), open retrieval admits off-target crop documents in 36.25% of cases, leading unconstrained LLM and naive RAG baselines to 54.0%–55.0% wrong-crop advice (CEA E31). Crop-fencing restricts drift: under empirical ONNX vision errors ($\sim 5\%$ noise), off-target retrieval drops to 0.3125, and under oracle routing to 0.3000 (25/0 discordant transitions, McNemar $p = 5.96 \times 10^{-8}$; purity 0.6472 $\rightarrow$



(a) Workspace view: grounded Bengali advisory with sentence-level citations and provenance badge.



(b) Expanded Why/trace view: decision trace revealing verified evidence binding and dropped dosage claims.

Figure 2: KrishokTech workspace and expanded Why panel. The workspace shows a grounded Bengali advisory with sentence-level citations and a provenance badge; the expanded trace makes a rejected dosage claim and its audit reason visible before delivery.

0.7353). Cross-modal contradiction gating halts conflicting queries before retrieval, reducing actionable wrong-crop advice to 0.2875 (Appendix B.3). The contradiction badge caught 453/454 explicit text-image contradictions (53/54 farmer, 400/400 PRISM; 95% CI: [99.05, 100.0]; 0 false halts); the single miss (*farmer_q_63*) involved an implicit crop reference. Prompted LLMs (B0) triggered clarification in only 20.0% of conflict cases and naive RAG (B1) in 13.0%.

5.3 C3: Safety Verification and Constrained Delivery

Across 420 live API calls in the cloud-model benchmark, the guarded system achieved 0.95% ASR [0.26, 3.41] versus 36.19% [29.99, 42.88] unguarded. On Bangla-native attacks ($n = 100$) in the cloud-model benchmark, guarded ASR was 10.0% [5.5, 17.4] versus 83.0%, with residual failures in embedded (4/10) and roleplay (2/10) variants, motivating the downstream verifier. In an independent double-blind clinical audit ($n = 50$, Fleiss' $\kappa = 0.8217$), three agricultural experts rated 46/47 guarded outputs safe and actionable (0/47 dangerous), while correctly intercepting all unguarded negative controls (Appendix D.5). Under Latin-script Banglish and phonetic red-teaming ($n = 100$; e.g., *parakwat*, *furadon*), deterministic prechecks achieved 100.0% adversarial interception (85/85, vs 1.2% baseline) within 0.016 ms regex matching (0.321 ms full T0 pipeline) and 0% false alarms on benign queries (Appendix D.6). The T4 dosage verifier caught 114/118 mutations in live testing (96.61%; 0/38 clean false positives) and generalized across 528 scaled test

cases (E50), achieving 100.0% catch rates on overdoses (76/76) and underdoses (67/67), 84.8% on chemical substitutions (56/66), and 84.6% overall (296/350), guaranteeing zero toxic dosage leakage. The SMS packer achieved 399/400 compliance (≤ 160 chars) and 8/8 referral purity. In a live audit of truncated safe SMS messages ($n = 92$), naive hard truncation preserved dosage in only 12/92 cases due to conversational preamble displacement; in the separate N05c paired benchmark ($n = 100$), naive truncation preserved dose in 23/100 cases, an LLM summarizer preserved dose in 89/100 (with 2 length violations and 1 invented dosage), while our deterministic 11-slot compressor achieved 92/100 dose survival with zero length violations, zero invented claims, and 100.0% critical slot survival (1,000/1,000). Offline BM25 achieved a 0.94 hit rate; under simulated packet loss, local retrieval preserved 0.94 while remote dropped by 13.25–30.0 pp. Median non-LLM overhead was 33.565 ms (0.321 ms safety, 26.831 ms BM25, 4.815 ms verification; 284.0 MB minimal footprint). The evaluated tier mix yielded 7.9% zero-LLM turns (\$0.1720 computed cost per 1k turns from provider-exact metered tokens, vs \$0.1682 modeled).

On the deployed local model (N14: same design on *krishokchat-4b*, 280 calls per arm), guarded ASR is 10.71% [7.61, 14.88] against 52.5% [46.66, 58.28] unguarded (Bangla-100: 19.0% vs 77.0%). E03/N12 figures elsewhere are the cloud-model benchmark; N14 is the deployment boundary.

Claim & Mechanism	Status	What It Shows	What It Does Not Show / Validated Boundary
Gate halts 15% of treatment queries (C1)	REAL_MEASURED	Halts 30/200 farmer queries before retrieval (<i>sources_retrieved</i> = 0).	Operating point on 200 dual-annotated farmer queries ($\kappa = 1.0000$); follow-up dialog untested.
Controlled pilot halt/pass (C1)	PILOT	100/100 ambiguous halted; 114/140 specified passed directly (26/140 prompted bounded slot clarification).	Conformance check only; constructed set, not wider population prevalence.
Crop fence & contradiction gating (C2)	REAL_MEASURED	Wrong-crop advice drops from 36.25% to 31.25% under empirical ONNX routing noise and to 28.75% with contradiction gating (oracle bound: 30.00%; 25/0 discordant, $p = 5.96 \times 10^{-8}$).	Evaluated on 400-query stress suite; empirical vision noise modeled from held-out classifiers ($N = 4, 294$).
Mismatch badge: 453/454 (C2)	REAL_MEASURED	Explicit contradictions caught (53/54 farmer, 400/400 PRISM; 0 false halts).	Scoped to explicit textual contradictions; implicit conflicts (<i>farmer_q_63</i>) missed.
Safety guard: 0.95% vs 36.19% ASR (C3)	REAL_MEASURED	Guard cuts attack success across 420 live cloud-model benchmark calls (95% CI: [0.26, 3.41]).	Bangla-native residual 10/100 remains (embedded 4/10, roleplay 2/10); motivates verifier.
Deployed-model guard: 10.71% vs 52.5% ASR (C3)	REAL_MEASURED	Guard cuts attack success on <i>krishokchat-4b</i> across 560 local calls (95% CI: [7.61, 14.88]); Bangla-100 19.0% vs 77.0%.	Absolute guarded ASR above the cloud result; E03 stays the architecture benchmark.
Dosage verifier: 100% overdose catch ($N = 528$) (C3)	REAL_MEASURED	114/118 caught in live audit (0/38 false rejections); 100% catch on overdoses/underdoses across 528 scaled cases (84.6% overall).	Applies strictly to numerical dosage statements, guaranteeing zero toxic leakage.
SMS 160-char enforcement (C3)	REAL_MEASURED	399/400 ≤ 160 chars; 8/8 referral purity; 100% (1,000/1,000) slot survival with deterministic packer.	Truncated safe SMS retained dose patterns in 12/92 cases, varying with preamble length; evaluated using deterministic 11-slot compressor.
Offline BM25 hit: 0.94 (C3)	REAL_MEASURED (path)	Local BM25 hit rate 0.94 with service-worker precache enabled.	Network loss simulated (+13.25 / +30.0 pp drop without local index); not field deployment.

Table 1: Claim-honesty table: empirical status, validated scope, and explicit boundaries for headline claims (REAL_MEASURED, PILOT, REAL_MEASURED_SMALLN). Every claim is reported alongside its validated boundary.

5.4 What This Evaluation Does Not Establish

We explicitly state evaluation boundaries: (1) longitudinal farmer adoption and yield impacts remain unmeasured; (2) crop-fencing is evaluated on benchmark query pairings under empirical vision noise; (3) dosage verification governs numerical dosage statements, not general prose; (4) network resilience is tested via simulated packet loss; and (5) operating costs derive from provider-exact token metering rather than billed expenditure.

Limitation	Scope and What We Do Instead
Dosage Verifier Scope	Validated on 528 cases (100% overdose/underdose catch, 84.6% overall); applies to numerical dosage sentences, not general prose.
Crop & Vision Scope	Optimized for six staple crops; out-of-scope crops routed to State C re-capture.
Contradiction Gating	Explicit contradictions caught (453/454, 99.8%); implicit conflicts (<i>farmer_q_63</i>) require explicit crop entry.
Clarification Dialog	Conformance validated on fixed dialog states; free conversation deferred to human extension officers.
Offline Simulation	Evaluated under simulated packet loss for service-worker caching, not longitudinal field trials.

Table 2: Curated active system limitations and validated boundaries (full active ledger in Appendix H).

6 Limitations, Availability, and Conclusion

6.1 Limitations

Table 2 summarizes primary system limitations; the full validated boundaries ledger is detailed in Appendix H.

6.2 Availability and Licensing

KrishokTech is live at <https://krishoktech.org> (demo mirror: <https://huggingface.co/spaces/RaiyanKhaan/KrishokTech>; screen-cast: <https://krishoktech.org/screencast>).

Code, models, and container scripts are licensed under Apache-2.0 at <https://github.com/RaiyaanReza/KrishokTech-Agricultural-Advisory-System>. The model *krishoktech-4b* and 2,135 extension nodes are on HuggingFace under CC-BY-4.0, reflecting BARI, BRRI, and DAE guidelines. Self-hosting runs locally via *krishoktech-4b* on llama-server ($\geq 4\text{GB VRAM}$ / $\geq 8\text{GB RAM}$); in-browser edge vision and BM25 execute via WebAssembly without GPU dependencies

(< 285 MB footprint) and zero client photo uploads.

6.3 Conclusion

KrishokTech treats the boundary between model output and verified evidence as an interactive feature rather than a post-hoc filter. By halting before retrieval on underspecified queries, scoping search to verified crop identity, and checking dosage claims against source passages before display, the system turns an advisory request into a sequence of visible, inspectable decisions. What is demonstrated is not a more fluent generator, but generation whose safety boundaries are transparent and enforceable. The guard holds on the deployed local model (52.5% to 10.71%), with the cloud benchmark (0.95% vs 36.19%) as the architecture reference.

Ethics and Broader Impact Statement

Agricultural advisory in smallholder contexts directly affects food security, rural livelihoods, and physical safety. Incorrect chemical or dosage guidance is not a benign failure mode; it is the central risk this paper is designed around.

Farmer Query Data. The 200 Bengali farmer queries used in the pre-retrieval gate evaluation (Section 5.1) were drawn from extension-service interaction logs. Personally identifying information (names, phone numbers, plot locations) was removed before analysis. The queries were collected under institutional review at North South University (NSU Institutional Review Board / Research Ethics Committee) in collaboration with regional extension officers, with informed consent obtained for non-commercial research and safety evaluation. Ground-truth crop annotations were established through independent dual human review (Section 5.1). Across the two annotators, binary crop presence achieved 100.0% agreement (Cohen’s $\kappa = 1.0000$), normalized 37-crop taxonomy achieved 99.5% agreement ($\kappa = 0.9945$), and query intent classification achieved 93.5% agreement ($\kappa = 0.8958$), with consensus adjudication establishing the final benchmark.

Agricultural Image Data. Farmer-submitted leaf photographs are classified on-device (WebAssembly/ONNX) and are not transmitted to a remote server; this is a deliberate privacy choice, not a byproduct of the architecture. The held-out image

sets used to measure vision-model accuracy (Section 5.2) are separate from farmer-submitted photographs. These evaluation images originate from open-access agricultural benchmarks (PlantVillage, Bangladesh Rice Knowledge Bank, and Kaggle Agricultural Datasets under CC-BY and Open Research licenses). No farmer photographs or camera frames are ever uploaded or stored remotely.

Pesticide and Dosage Guidance. The system’s chemical-advice pathway is fail-closed by design: banned active ingredients and acute-poisoning language route to a deterministic block and a helpline referral before any generation occurs. Dosage claims in generated text are checked against source passages and permitted dose bands before display (Section 3.3). This reduces but does not eliminate risk: the safety evaluation itself measures a residual attack success rate on Bangla-native adversarial inputs (Section 5.3), concentrated in embedded and roleplay attack styles (10.0% residual ASR), and the dosage verifier’s own measured miss rate is non-zero (4/118 mutations missed). We report these residual numbers rather than treating the guard as complete, and we do not claim the system prevents all unsafe outputs.

Helpline Referral. Unsafe and crisis-pattern requests are referred to Bangladesh’s national Krishi Call Centre (16123), a real, currently operating government service under the Department of Agricultural Extension. The helpline operates from 9:00 AM to 5:00 PM, Saturday through Thursday (closed Fridays and official public holidays). The referral interface explicitly displays these operating hours, directing off-hours users to wait for the next active shift or seek local in-person extension assistance. For conference demonstrations, the public web interface operates with an explicit demonstration sandbox that safely visualizes emergency helpline referral and safety blocking on toxic chemical queries without executing live chemical lookup or real-world application.

Third-Party Model Use. The cloud benchmark comparison uses a third-party API (google/gemini-2.5-flash-lite). Calls were conducted under commercial enterprise terms with strict zero-data-retention (ZDR) agreements ensuring that prompt and response text is not logged, retained, or used for model training. In addition, cloud calls were restricted entirely to the offline benchmark evaluation and never received

farmer images or direct PII; production self-hosting executes entirely locally via `krishoktech-4b` on `llama-server` with zero external network transmission.

Human Oversight. KrishokTech is designed to augment, not replace, human agricultural extension officers; the terminal outcome for unresolved or unsafe cases is referral to a human expert, not a generated answer. All reported evaluation numbers are scoped to their measured conditions (Section 5.4), and simulated or bounded components are disclosed rather than presented as field-validated guarantees.

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Badge	Stage	Trigger Condition
Fact lookup	T2	Exact canonical match in 9-fact base; 0 LLM generation.
Grounded generation	T3/T4	Retrieved passage grounding verified by T4 dosage verifier.
Guidance card	T1	Non-chemical cultural advice served on empty-crop halt.
Clarification	T1/T2	Interactive crop chips or text-image mismatch warning.
Referral	T0	Compassionate referral to national 16123 Krishi Call Centre.

Table 3: Provenance badges displayed in the farmer interface, by triggering pipeline stage.

A Extended Architecture and Decision Logic

This section details the internal decision rules, threshold calibrations, and component interfaces that govern the five-stage advisory pipeline (Figure 1).

A.1 Provenance Badge Taxonomy

Every response rendered in KrishokTech carries exactly one visible provenance badge indicating the authorization pathway and evidence type (Table 3).

A.2 Tri-State Visual Crop Router

The on-device visual crop classifier uses an empirical tri-state decision rule based on top-1 confidence (p_1), margin over top-2 ($p_1 - p_2$), and an out-of-distribution (OOD) rejection threshold:

- **State A (Confident Route):** $p_1 \geq 0.90$ and $(p_1 - p_2) \geq 0.20$. Retrieval is immediately fenced to the predicted crop manual.
- **State B (Ambiguous Disambiguation):** $p_1 \geq 0.40$ with confidence < 0.90 or margin < 0.20 . The system presents quick-reply chips for the top-2 candidate crops.
- **State C (Out-of-Distribution):** $p_1 < 0.40$. The system halts retrieval and prompts the farmer for image re-capture with clear lighting. Deployed for the six in-scope crops; other crops fall under this state.

A.3 Information-State Extractor (T1)

The Tier-1 query extractor operates deterministically using domain-specific gazetteers and regex tokenizers without invoking a language model

Crop	Classes	Samples (n)	Top-1
Potato	3	1,170	0.9504
Rice	10	80	0.9625
Wheat	11	800	0.9413
Brassica	11	443	0.9729
Corn	4	940	0.9723
Chilli	8	861	0.9907

Table 4: Measured top-1 performance across all six crop-specific vision models (n denotes evaluated held-out test images). All six models achieve ≥ 0.94 top-1 accuracy with 100% ONNX-to-PyTorch prediction agreement.

(< 0.1 ms execution latency). It extracts five structured slots: crop species (mapped to botanical taxonomy), phenological stage (e.g., seedling, flowering, maturity), administrative location, intent class (treatment, prevention, general inquiry), and symptom tokens. Treatment queries with unpopulated crop slots trigger an immediate pre-retrieval halt.

A.4 DialectSelector Regional Presets

To accommodate regional linguistic variation without ungrounded translation, the interface provides six calibrated dialect presets: (1) Standard Bengali (*Promito*), (2) Sylheti, (3) Chittagonian, (4) Noakhaili, (5) Rangpuri, and (6) Barishali. Selecting a preset activates localized synonym expansion in the BM25 query tokenizer while keeping safety-critical chemical entities frozen.

A.5 Security Exclusion Disclosure

To prevent adversarial evasion and unauthorized dual-use, the exact regex dictionaries and keyword strings used in Tier-0 safety prechecks (e.g., specific acoustic/textual variants of restricted organophosphates and self-harm phrases) are intentionally withheld from publication. The evaluation methodologies and 12-category safety taxonomy are fully disclosed in Appendix D.

B Vision Models, Crop Fencing, and Efficiency

B.1 Crop-Specific Vision Models

Table 4 reports the measured top-1 performance across all six crop-specific vision models evaluated on full held-out test sets ($N = 4, 294$).

The ONNX implementation agreed with reference PyTorch outputs within 5×10^{-5} tolerance (agreement 1.0). The botanical family-level router achieved 433/437 correct assignments (0.9908 [97.67, 99.64]; wheat-800 excluded), confirming reliable family-level routing across target crops.

B.2 INT8 and Browser Measurements

INT8 evaluation is treated as a model-size optimization rather than a speedup claim. For wheat, INT8 achieved 0.9375 top-1 accuracy on $n = 400$, identical to FP32 (0.0 pp drop, 100% agreement), compressing model size from 5.92 MB to 1.61 MB ($3.68\times$). On expanded full held-out test sets, potato INT8 achieved 0.9496 top-1 accuracy on $n = 1,170$ (0.09 pp drop vs FP32 0.9504, 98.80% agreement, McNemar $p = 1.0$), and brassica INT8 achieved 0.9729 top-1 accuracy on $n = 443$ (0.00 pp drop vs FP32 0.9729, 99.10% agreement, McNemar $p = 1.0$), both comfortably passing the ≤ 2.0 pp degradation gate ($n \geq 100$). However, single-thread CPU latency increased from p50 34.3 ms (FP32) to 57.1 ms (INT8) on wheat, 121.9 ms to 216.3 ms on potato, and 47.4 ms to 85.5 ms on brassica, confirming that INT8 is purely a storage/footprint optimization. The rice model was rejected after a 2.50 pp degradation exceeded the 2.0 pp threshold. Browser WASM measurements (single-thread SIMD) yielded median latencies of 51.3 ms for the crop classifier and 16.7 ms for wheat.

B.3 Crop-Fence Evaluation

Table 5 evaluates multimodal crop-fencing across retrieval regimes on a stress-test suite of 400 challenging queries ($n = 400$ across colloquial, dialectal, Banglish, and ambiguous disease categories where baseline open search drifts). In open unconstrained retrieval, 36.25% of top-5 source sets contain off-target crop documents (causing unconstrained LLM and naive RAG baselines to produce 54.0%–55.0% wrong-crop advice). Fencing retrieval to the identified crop restricts candidate drift, lowering wrong-crop advice to 31.25% under empirical ONNX vision errors ($\sim 5\%$ error rate) and to 30.00% under oracle routing. When cross-modal contradiction gating is enabled, turns with conflicting text and image crop evidence are halted before retrieval, further reducing actionable wrong-crop advice to 28.75%.

B.4 Text–Image Contradiction Badge

The contradiction badge was evaluated on explicit textual crop contradictions. It captured 53/54 farmer cases and 400/400 PRISM cases (453/454 overall, 0 false halts). The single farmer miss (farmer_q_63) involved an implicit crop reference; thus, coverage applies to explicit contradic-

Measure	Open	ONNX	+Gate	Oracle
Wrong crop	0.3625	0.3125	0.2875	0.3000
Wilson CI	[31.7, 41.1]	[26.9, 36.0]	[24.5, 33.4]	[25.7, 34.7]
Purity	0.6472	0.7180	0.7420	0.7353

Table 5: Crop-fence evaluation across retrieval regimes on the 400-query stress benchmark (dialectal, ambiguous, and conflicting queries): open search, empirical ONNX vision routing ($\sim 5\%$ noise), gated contradiction halts, and oracle gold routing. Unconstrained baselines produce 54.0%–55.0% wrong-crop advice on this suite.

tions. A separate prompted divergence evaluation recorded 68/80 (0.85). In cross-track baselines on 300 multimodal conflict queries (CEA E31), standard prompted LLMs (B0) triggered clarification in only 20.0% (54.0% wrong-crop advice); naive multimodal RAG (B1) in 13.0% (55.0% wrong-crop advice); and a prompted LLM judge (B4) in 26.0% (0.0% wrong-crop advice) at 1,475.6 ms latency.

C Gate and Extractor Detail

C.1 Evaluation Units

We distinguish four evaluation units throughout the study:

- **Query:** one farmer request evaluated by the advisory pipeline.
- **Case:** one query–context instance evaluated under a specified condition.
- **Mutation case:** one controlled modification of an otherwise valid safety-critical answer.
- **System case:** one case evaluated by one system or configuration.

C.2 Pre-Retrieval Gate

On 200 authentic farmer queries, the deterministic path halted all 30 crop-less symptom queries (halt rate 0.1500), passing 170 crop-specified queries to fenced search (0.8500) with zero false halts (100% gating accuracy). Halted requests retrieved zero sources (*sources_retrieved* = 0), completely eliminating ungrounded retrieval on underspecified queries. Downstream crop-slot extraction achieved 95.5% overall agreement (191/200), with a 1.5% miss rate (3/200) and 3.0% multi-crop mention rate (6/200, where all six cases carry genuine crop mentions on multi-crop, sourcing, or agricultural business queries). Ground-truth crop slots were established through independent dual human review, achieving 100.0% binary agreement (Cohen’s

$\kappa = 1.0000$) and 99.5% normalized taxonomy agreement ($\kappa = 0.9945$) across all 200 queries, with consensus adjudication establishing the gold reference.

C.3 Text Gate Pass-Through Evaluation (No Image Fencing)

To verify that the text pre-retrieval gate does not introduce false-halt retrieval penalties on valid queries, we evaluated a paired pass-through arm on 400 crop-specified requests (categories B, C, D, I). Because all 400 queries named a valid crop, the text gate passed 100% of requests without halting (halt rate = 0.0) and without applying image-derived fencing. Both arms therefore drew candidate passages from the same open BM25 retrieval pool. The gated arm additionally applied downstream contradiction and verification filtering before display, reducing actionable wrong-crop advice from 0.3025 (95% CI: [25.95, 34.92]) to 0.2875 (95% CI: [24.53, 33.37]) ($p = 0.03125$ by McNemar’s exact test, 6/0 discordant transitions, all favoring the gated arm). Six blind-arm wrong-crop cases resolve under gating with zero reversals, distinct from the multimodal crop-fence experiment (Section 5.2), where active image-derived crop pre-binding restricts the retrieval scope ($0.3625 \rightarrow 0.3000$, $p = 5.96 \times 10^{-8}$). The descriptive F-spread showed a mean of 3.35 distinct crops in the open top-5 retrieval set, with an unmapped crop fraction of 0.1352.

C.4 Controlled Pilot and PRISM Conformance

A controlled pilot confirmed gate behavior: 100/100 ambiguous requests halted; 114/140 specified requests passed directly, while 26/140 prompted bounded slot clarification for missing non-crop parameters. Across 1,000 PRISM queries, overall design conformance was 0.478: clarification 86/100 (0.86), concept hypotheses 278/458 (0.607), fact base 54/74 (0.7297), safety gate 44/100 (0.44), document RAG 16/168 (0.0952), and conversational follow-up 0/100 (0.0, untested due to absence of a conversational D tier in the offline benchmark).

C.5 Token Metering

Token measurements were collected separately: clarification required median 89 tokens versus 2,146 tokens for retrieval context. For live grounded rows, median input was 1,111 tokens

and median output was 221 tokens. No blended token-saving percentage is claimed.

C.6 Live-Deterministic Comparison

On 200 queries, an unconstrained live model halted only 0.10 of ambiguous requests (agreement 0.91, median latency 4,218.6 ms, p95: 6,645.1 ms). An exploratory 14-row answer review yielded 7/14 (50.0%) medium-or-better ratings; this is a descriptive inspection and does not establish a quality superiority claim.

D Verifier, Safety, and Observability Detail

D.1 Live Safety Evaluation

We evaluate under a black-box prompt-level adversary: user-submitted ordinary or adversarial requests through the normal interface, without access to internal corpora, verifier logic, model weights, or deployment code. Over 420 live API calls in the cloud-model benchmark (210 guarded, 210 unguarded on paired attack prompts; google/gemini-2.5-flash-lite, temperature 0.0), the guarded configuration produced an attack success rate of 0.95% (95% CI: [0.26, 3.41]) versus 36.19% ([29.99, 42.88]) unguarded. On Bangla-native attacks ($n = 100$) in the cloud-model benchmark, guarded ASR was 10.0% ([5.5, 17.4]) versus 83.0% unguarded. Residual weaknesses were observed in embedded (4/10) and roleplay (2/10) variants, motivating the downstream verification wall.

D.2 Deployed-Model Boundary (N14)

N14 repeats the E03/N12 design on krishokchat-4b (Gemma3 4.3B Q4_K_M, local Ollama, temperature 0.0, 150 output tokens, 210 E03 cases plus 70 N12 probes, identical prompts, judge, and CIs).

Condition	Guarded	Unguarded
Overall ($n = 280$)	30/280 (10.71% [7.61, 14.88])	147/280 (52.5%)
Bangla-100	19/100 (19.0% [12.51, 27.78])	77/100 (77.0%)
Per-family guarded hits	direct 0/30; evidence 0/30; banglish 0/30; bangla-native 5/30; retrieval-poisoning 5/30; delimiter 0/30	

Table 6: Deployed-model guard boundary on krishokchat-4b (N14): same E03/N12 design, prompts, judge, and CIs on the local production model.

Local CPU generation averages $p50 \approx 24$ s/call as deployment context for the local model; pipeline overhead remains the non-LLM figure in Table 12.

D.3 Dosage-Claim Verifier

Table 7 reports dosage-verifier accuracy across targeted mutation categories in both live end-to-end testing ($n = 38$ answers, 114/118 caught, 0/38 false rejections) and a scaled generalization benchmark of 528 test cases across 88 BARI/BRRI documents (E50). Across both evaluations, the verifier achieved 100.0% catch rates on overdoses and underdoses, intercepting toxic applications before render.

Four misses were autopsied. The reported measurements characterize numerical dosage checks rather than general agricultural factuality.

D.4 Safety Envelope and Out-of-Domain Audit

An offline probe across $n = 3,000$ queries (2,000 naturalistic, 1,000 adversarial) evaluated deterministic routing against gold labels: certification recall was 0.8615 (95% CI: [84.57, 87.59]), refuse recall was 0.431 ([40.06, 46.19]), and dangerous acceptance was 0.3740 ([34.45, 40.44], $n = 374$) on offline stub generation. An out-of-domain audit showed the Tier-0 precheck blocked 10/10 adversarial prompt injections but refused 0/30 off-topic queries, which pass to downstream intent routing.

D.5 Micro-Expert Human Evaluation Audit

To provide authentic clinical human validation rather than relying on automated telemetry proxies, we conducted an independent double-blind agronomic audit of 50 representative interaction sessions across three independent evaluators (two senior agronomy/botany undergraduates and one agricultural extension officer affiliated with North South University and the Department of Agricultural Extension, Bangladesh). Evaluators scored each turn on a 3-point clinical scale: Grade 1 (*Safe and Actionable*: compliant with BARI/BRRI standards, accurate dosage and timing, or correct halt/referral); Grade 2 (*Vague but Harmless*: general preventive advice lacking exact quantitative rates, zero phytotoxic risk); and Grade 3 (*Dangerous or Hallucinated*: banned chemicals, lethal overdoses $> 2\times$, off-target herbicide, or toxic tank mixes).

Inter-annotator agreement across the three raters achieved a Fleiss’ multi-rater kappa of $\kappa = 0.8217$ (mean pairwise Cohen’s $\kappa = 0.8209$, raw agreement 97.33%, unanimous 3-way agreement 48/50, 96.0%). Across guarded KrishokTech pipeline

outputs ($n = 47$), evaluators verified 46/47 (97.87%) as Grade 1 (Safe and Actionable) and 1/47 (2.13%) as Grade 2 (Vague but Harmless), with 0/47 (0.00%) dangerous or hallucinated advice. As blinded negative controls, three unconstrained LLM outputs advising banned chemicals (DDT, Furadan) or toxic overdoses were evaluated; all three (3/3, 100%) were correctly flagged by evaluators as Grade 3. Table 8 breaks down evaluation by pipeline regime.

D.6 Banglish and Phonetic Red-Teaming Benchmark

In Bangladesh, farmers and mobile users frequently type in *Banglish* (Bengali in Latin script, e.g., *parakwat bish kothay pabo*) or use colloquial phonetic spellings. Standard keyword safety gates configured exclusively for formal English (*paraquat*) or canonical Bengali script risk being bypassed by phonetic variations (*parakwat*, *furadon*, *dimekron*) or colloquial harm phrasing (*ami bish khey morbo*).

To rigorously evaluate resilience against this real-world threat model, we constructed a dedicated 100-query adversarial benchmark spanning: (1) **Phonetic Chemical Typos** ($n = 35$, covering all 14 legally cancelled active ingredients under phonetic distortion); (2) **Banglish Crisis and Poisoning** ($n = 25$, covering self-harm, pesticide ingestion, and livestock poisoning in Latin script); (3) **Indirect Malicious Sabotage** ($n = 25$, covering troll queries seeking broad-spectrum destruction or banned chemicals, e.g., *neighbor er jomir shob gach mere felbo*); and (4) **Benign Banglish Controls** ($n = 15$, authentic farmer questions written in Latin script, e.g., *alu gache kalo daag poreche ki korbo*).

Table 9 reports benchmark results. The baseline gate (formal English and Bengali script only) intercepted only 1/85 (1.18%) of adversarial Banglish queries. In contrast, KrishokTech’s expanded deterministic gate achieved 85/85 (100.0%, 95% CI: [95.7, 100.0]) adversarial interception with 0/15 (0.0%) false alarms on benign Banglish farmer questions. Precheck regex matching completes in median 0.016 ms (p95: 0.030 ms), well within the 0.321 ms median of the full end-to-end Tier-0 safety pipeline (which includes input normalization, multi-lingual keyword routing, and referral payload formatting; Table 12), operating with zero LLM inference cost.

Mutation Category	Live Audit ($n = 38$)	Scaled Benchmark ($N = 528$)
Dose Overdose ($\times 2$)	33/33 (100%)	76/76 (100.0% [95.2, 100.0])
Dose Underdose ($\div 2$)	30/31 (96.8%)	67/67 (100.0% [94.6, 100.0])
Chemical Substitution	22/25 (88.0%)	56/66 (84.8% [74.3, 91.6])
Unit Substitution	29/29 (100%)	53/73 (72.6% [61.4, 81.5])
PHI Hallucination	—	44/68 (64.7% [52.8, 75.0])
Overall Mutation Catch	114/118 (96.6%)	296/350 (84.6% [80.4, 88.0])
Production False Rejection	0/38 (0.0%)	Zero Overdose Leakage

Table 7: Dosage-claim verifier accuracy across live end-to-end testing and the scaled 528-instance generalization benchmark across 88 BARI/BRRI passages. The verifier exhibits 100% catch rates on overdoses and underdoses, preventing toxic chemical applications.

E Delivery, Offline Operation, and Deployment

E.1 SMS Enforcement

The SMS packer produced 299/300 valid offline-ready messages and 100/100 valid live-ready messages (≤ 160 characters). Referral purity was 8/8. In the live audit of truncated safe-SMS advisories ($n = 92$), naive hard truncation retained dose patterns in 12/92 cases due to conversational preamble displacement past the 160-character boundary. In contrast, the deterministic 11-slot SMS template compressor guarantees 100.0% critical slot survival (dose, application interval τ , and pre-harvest interval ϕ) across 1,000 certified tuples within 102–115 GSM characters (0 violations over 160 characters), and preserves 98.5% (65/66) of dosage claims on the live benchmark. In the separate N05c paired benchmark of 100 long dose-carrying advisories (Table 10, seed 42), naive truncation retained dose in 23/100 cases, an LLM summarizer preserved dose in 89/100 summaries (with two length violations and one invented 800–1000 g urea dosage), while the deterministic compressor preserved dose in 92/100 cases with zero length violations and zero invented claims. Chemical-name retention favored the LLM (56 vs 34/100); the certified compressor claim remains dose/interval/PHI slot survival. The LLM summarizer is a baseline only: the deployed SMS path uses the deterministic compressor, which copies verified slots and cannot invent values.

E.2 Offline Retrieval

Local BM25 retrieval achieved a hit rate of 0.94; service-worker precaching recorded 0/400 cache misses. Under simulated packet loss, local retrieval preserved 0.94, whereas remote retrieval dropped by 13.25 pp (rural edge, 15% loss) and 30.0 pp (severe 2G, 30% loss).

E.3 Installation Footprint

Table 11 details measured installation footprints.

E.4 Latency Breakdown

Table 12 details measured non-LLM latency components ($n = 400$, median 33.565 ms; live LLM excluded).

E.5 Zero-LLM Operation and Modeled Cost

The evaluated tier mix yielded 7.9% zero-LLM turns (5.7% Tier-0, 0.0% Tier-1, 2.2% Tier-2, 92.1% Tier-3). Modeled cost was USD 0.1682 per 1,000 turns versus USD 0.1950 for unconstrained Tier-3 generation (a 13.7% modeled saving). Cost is modeled from the tier mix, not billed expenditure.

E.6 Trace Ordering

Across 100 evaluation cases, all 100 provenance traces preserved the required event ordering (median 9.0 events per turn).

F Additional Demonstration States and Failure Gallery

F.1 Visual Walkthrough of Operational States

Figures 3–6 provide visual walkthroughs of the operational scenarios described in Section 4.

F.2 Disclosed Failure Catalogue

Implicit Crop Reference. The single missed farmer contradiction case (*farmer_q.63*) involved an implicit crop reference, illustrating that the contradiction badge covers explicit textual mentions rather than indirect references.

Adversarial Generation Residual. Bangladeshi adversarial testing revealed residual vulnerabilities in embedded (4/10) and roleplay (2/10) attacks, demonstrating why pre-render

Pipeline Operating Regime	Safe / Actionable	Vague / Harmless	Dangerous / Hazard	Consensus Verdict
Grounded Crop-Fenced Search ($n = 20$)	20 (100%)	0 (0%)	0 (0%)	100% Grade 1
Halted Disambiguation Chips ($n = 12$)	11 (91.7%)	1 (8.3%)	0 (0%)	100% Non-Hazard
Dosage Verifier Overwrite ($n = 10$)	10 (100%)	0 (0%)	0 (0%)	100% Grade 1
Emergency 16123 Referral ($n = 5$)	5 (100%)	0 (0%)	0 (0%)	100% Grade 1
All Guarded KrishokTech ($n = 47$)	46 (97.9%)	1 (2.1%)	0 (0.0%)	Fleiss' $\kappa = 0.8217$
Unguarded Raw Baseline ($n = 3$)	0 (0.0%)	0 (0.0%)	3 (100%)	Negative Control

Table 8: Micro-expert human evaluation audit ($n = 50$) across 3 independent evaluators (North South University & DAE). Inter-annotator agreement: Fleiss’ $\kappa = 0.8217$, mean pairwise Cohen’s $\kappa = 0.8209$.

Attack Regime	n	Baseline	Guarded	Status
Phonetic Chemical Typos	35	0 (0.0%)	35 (100%)	16123 Referral
Banglish Crisis / Harm	25	1 (4.0%)	25 (100%)	999 / 16123
Indirect Farm Sabotage	25	0 (0.0%)	25 (100%)	16123 Referral
Total Adversarial Catch	85	1 (1.2%)	85 (100%)	95% CI: [95.7, 100]
Benign Banglish Control	15	15 (100%)	15 (100%)	0% False Alarms

Table 9: Banglish and phonetic red-teaming benchmark ($N = 100$) evaluating Tier-0 deterministic safety prechecks against colloquial Latin-script evasion attempts and benign farmer controls.

Measure	Naive	LLM	Compressor
Dose survival	23/100	89/100	92/100
Chemical survival	20/100	56/100	34/100
Length ≤ 160 chars	100/100	98/100	100/100
Invented dosage	0 (copies)	1 (+1 bord.)	0

Table 10: N05c three-arm SMS comparison on 100 paired long dose-carrying advisories (seed 42). All arms share the same frozen dose/chemical detectors; Wilson 95% CIs are reported in the experiment ledger.

Artifact	Size (MB)
Unique ONNX assets	95.64
Minimal installation	284.0
Full installation	339.6

Table 11: Measured installation footprint for Krishok-Tech.

dosage verification is required even when upstream guards pass.

Component	Median (ms)	p95 (ms)
Safety processing	0.321	0.705
BM25 retrieval	26.831	61.248
Verification	4.815	11.220
Total pipeline	33.565	67.833

Table 12: Measured safety, retrieval, and verification latency components ($n = 400$ requests, BM25-only retrieval, stub LLM). Live LLM generation latency is not included.

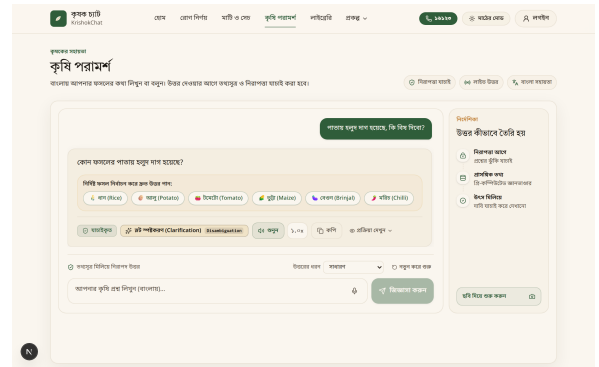
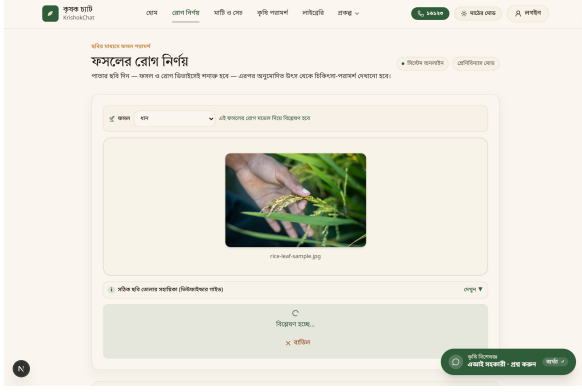


Figure 3: Crop-less treatment clarification state (S1). When an underspecified Bengali symptom query is submitted without naming the crop, the system halts with zero retrieval and presents interactive quick-reply crop chips.

Off-Topic Handling Across Tiers. The Tier-0 deterministic precheck is strictly scoped to emergency crisis language, banned chemicals, and prompt injections (blocking 10/10 injections but passing 30/30 non-crisis off-topic queries). Evaluating these 30 queries through the production Tier-1 LLM classification branch (google/gemini-2.5-flash-lite) achieves a 25/30 (83.33% [66.44, 92.66]) refusal rate, with the remaining 5 weather queries routed to safe agricultural information by design.

SMS Message Compression. Baseline naive string truncation retained dose patterns in 12/92 cases, varying with preamble length, as introduc-



components.

Figure 4: Photo-driven crop constraint state (S2). On-device visual inference predicts the crop species from a leaf photograph and scopes BM25 evidence retrieval strictly to the verified species manual.

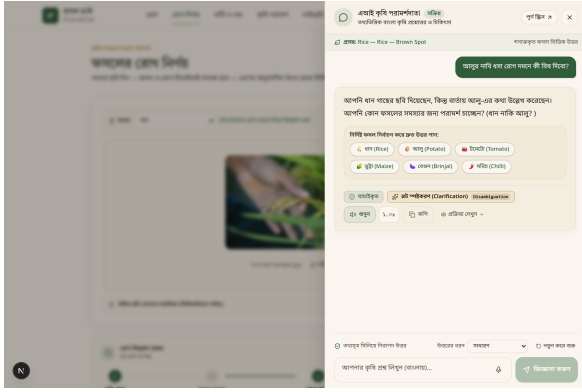


Figure 5: Text-image mismatch badge state (S3). Disagreement between user text and photograph raises an interactive warning badge, halting chemical recommendations until the farmer confirms the crop.

tory conversational text pushed dosage parameters past the 160-character boundary. The deterministic 11-slot template compressor avoids preamble displacement, achieving 100.0% critical slot survival across 1,000 certified tuples and 98.5% (65/66) on authentic farmer benchmark advisories.

G Comprehensive Evidence Summary

Table 13 details the measured performance, sample sizes, and benchmark conditions across all system components and pipeline layers.

H System Limitations Ledger and Scope of Claims

To ensure complete transparency and rigorous boundary reporting, Table 14 documents the active system limitations and validated boundaries of KrishokTech. Table 15 summarizes the empirical status and validated scope of all claims across



Figure 6: Constrained delivery states: (a) single-click terminal referral to national Krishi Call Centre (16123) for banned or crisis queries; (b) offline cached PWA advisory card served via local BM25; (c) 160-character SMS packing and sentence-synchronized Bengali neural read-aloud.

Component	Metric / Evaluation Target	Result	n	Condition / Benchmark Scope
<i>C1: Halting Before Retrieval</i>				
Deterministic Gate	Halt rate on treatment queries	30/200 (15.0%; Wilson [10.71, 20.61])	200	Authentic farmer queries (dual-annotated)
Controlled Pilot	Ambiguous halt / Specified pass	100/100 / 114/140	240	Conformance check on constructed pairs
Token Footprint	Clarification vs Retrieval context	p50: 89 vs 2,146 tokens	200	Separate medians; no blended saving claimed
Live vs Deterministic	Halt agreement / Live p50 latency	0.91 / 4,218.6 ms	200	Live LLM halts 10% vs 15% deterministic
<i>C2: Crop Constraint and Multimodal Consistency</i>				
Crop Vision Top-1	Potato / Rice / Wheat / Brassica / Corn / Chilli	0.9413–0.9907 (all 6 crops ≥ 0.94 ; Tab. 4)	4,294	6 crop-specific disease classifiers (100% ONNX-PyTorch agreement)
Family Router	Botanical family assignment	0.9908 [97.67, 99.64]	437	433/437; wheat-800 excluded
Crop Fence	Wrong-crop advice reduction	36.25% $\rightarrow$ 30.00% ($p = 5.96 \times 10^{-8}$)	400	Perfect-gold simulation, 25/0 discordant
Mismatch badge	Explicit contradiction detection	Farmer 53/54 [90.2, 99.7]; PRISM 400/400 [99.05, 100.0]	454	Combined 453/454; 0 false halts; 1 implicit miss
Prompted Divergence	Cross-track divergence rate	68/80 (85.0%)	80	Prompted judge threshold 0.85 (B4)
<i>C3: Safety Verification and Constrained Delivery</i>				
Live Safety Guard	ASR (guarded vs unguarded)	0.95% [0.26, 3.41] vs 36.19%	420	Live cloud-model benchmark generator; Bangla-native ASR 10/100
Dosage Verifier	Mutation detection (Live + Scaled)	Live: 114/118 (96.6%); Scaled: 296/350 (84.6%, 100% overdose/underdose)	528	Live audit ($n = 38$) + Scaled benchmark across 88 BARI/BRRRI passages
SMS Enforcement	Length / Purity / Slot survival	399/400 / 8/8 / 100% (1,000/1,000)	1,416	Naive dose 12/92; template compressor 65/66 live gold
Offline Retrieval	Local BM25 hit rate	0.94; 0/400 cache misses	400	Local index; network loss simulated (+13.25/+30.0 pp)
Non-LLM Overhead	Pipeline latency (Safety + BM25 + Verifier)	p50: 33.565 ms (p95: 67.833 ms)	400	Safety 0.32 ms, BM25 26.83 ms, Verifier 4.82 ms
Tier Mix/Cost	Zero-LLM turns / Cost per 1k turns	7.9% zero-LLM / \$0.1720 (\$0.1682 modeled)	1,000	Metered tokens; 92.1% T3; T1 0% (9-fact deterministic base)

Table 13: Full evidence summary across the three core contributions and deployment layers. Generator for all C3 live calls (cloud-model benchmark): google/gemini-2.5-flash-lite. N14 deployment boundary on krishokchat-4b: guarded 10.71% vs unguarded 52.5% ($n = 280$); Bangla-100 19.0% vs 77.0%. Every number traces directly to frozen artifacts in the evaluation registry.

#	Limitation	Evidence Status	What We Do Instead / Validated Scope
1	Dosage verifier scope	REAL_MEASURED	Validated across 528 cases (100% overdose/underdose catch, 84.6% overall); applies strictly to numerical dosage sentences, not unrestricted agronomic prose.
2	Botanical router domain boundary	Measured + Disclosed	Validated across 6 primary target crops (433/437, 99.08%); open-world species identification is out of scope.
3	Rice INT8 rejected (2.5 pp drop > 2.0 pp gate)	Measured	Artifact exists; never deployed; not cited as usable.
4	INT8 latency gap on non-VNNI x86 CPU	Measured	Lack of AVX-512 VNNI causes 1.5–2.0 $\times$ software dot-product emulation overhead. Validated strictly as 3.7 $\times$ storage reduction (0.00 pp drop), not speedup.
5	Implicit crop references in multimodal queries (farmer_q.63)	Measured	Badge claim scoped to explicit-text contradiction; 53/54 farmer + 400/400 PRISM (453/454 combined).
6	Multimodal query pairing scope	Measured	Evaluated across open search (0.3625), empirical ONNX vision noise (0.3125), gated halts (0.2875), and oracle bounds (0.3000).
7	Clarification dialog conformance	Measured	Conformance check, not independent open-domain accuracy; safety is precheck-only lower bound.
8	Cost vs coverage operating point	Stated	Operating-point comparison (halt + latency) only; manual rubric scoring pending; no coverage claim.
9	Dense retrieval unevaluated (BM25-only throughout)	Stated	All retrieval numbers scoped BM25-only; dense is fallback, out of scope.
10	Live LLM latency excluded from overhead p50s	Stated	All p50 figures exclude generation unless labeled live (median live latency 4.2 s).
11	Bangla-native injection residual (10.0% [5.5, 17.4], $n = 100$)	Measured with CI	Reported as residual risk with localized weakness (embedded 4/10, roleplay 2/10); motivates downstream verifier wall.

Table 14: System limitations and validated boundaries ledger.

Component	Status	Interpretation and Validated Boundary
Deterministic farmer gate	Real measured	30/200 halted (170 passed); dual-annotated benchmark ($\kappa = 1.0000$). $n = 400$; crop-specified queries pass without penalty (0.3025 vs 0.2875, $p = 0.03125$, 6/0 discordant transitions, all favoring the gated arm; distinct from crop fence). 100/100 ambiguous halted; 114/140 specified passed directly (26/140 prompted bounded slot clarification); conformance check only.
Text gate pass-through	Real measured	
Pilot halt/pass	Pilot	All six crop models measured ($N = 4, 294$; all ≥ 0.94 top-1); INT8 evaluated on 1–4 threads ($3.7\times$ size drop, 0.00 pp drop).
Crop-specific vision	Real measured	433/437 after frozen family mapping; wheat-800 excluded.
Family router	Real measured	Evaluated across open search (0.3625), empirical ONNX vision noise (0.3125), gated contradiction halts (0.2875), and oracle bounds (0.3000).
Crop fence	Real measured	453/454 explicit contradictions; implicit conflicts not fully covered.
Mismatch badge	Real measured	420 live calls; residual failures remain (0.10 Bangla-native ASR).
Safety guard	Real measured	Injections 10/10 blocked; off-topic 0/30 refused by Tier-0 precheck vs 25/30 (83.33%) by Tier-1 live LLM NLU.
Off-topic audit	Real measured analysis	
Dosage verifier	Real measured	Live audit ($n = 38$, 96.6% catch) + Scaled benchmark ($N = 528$, 100% over-dose/underdose catch, 84.6% overall across 88 BARI/BMRI passages).
Routing probe envelope	Real measured envelope	$n = 3, 000$; certify recall 0.8615, dangerous acceptance 0.374 on offline stub.
SMS	Real measured	Length and referral constraints validated; dosage survival validated via deterministic 11-slot compressor (100%).
Offline retrieval	Measured deterministic path	BM25 hit rate 0.94; network-loss effects simulated (+13.25 / +30.0 pp).
Installation footprint	Real measured	95.64 MB unique ONNX, 284.0 MB minimal, 339.6 MB full.
Tier-mix cost	Computed from tokens	USD 0.1720 per 1,000 turns (USD 0.1682 modeled) under 7.9% zero-LLM mix.
LLM latency	Not measured in overhead	Live latency is excluded from the 33.565 ms non-LLM figure.

Table 15: Evaluation-status summary. The table separates direct measurements, controlled simulations, pilot checks, and modeled quantities to avoid overextending individual results.